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Intermittent Controllable Magnetically Levitated Turbine Generator System in Water Injection Wells

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Abstract

To address the critical challenge of sustainable power supply for downhole wireless sensor networks in intelligent oilfields, this study proposes an intermittently controllable magnetically levitated turbine generator system specifically designed for water injection wells. Targeting limitations of conventional turbine generators—including mechanical wear, low-flow-rate sensitivity, and spatial constraints in narrow wellbores—the system integrates three key innovations: (1) A magnetically levitated turbine design where the impeller and rotor are suspended via magnetic coupling, significantly reducing rotational friction and enabling a low startup flow rate of 24.5 m³/d; (2) Intermittent power generation control using an eccentrically positioned flow path selector valve to direct fluid between a central bypass channel and a power generation channel, supporting on-demand operation; (3) Intelligent energy management where a main control chip monitors battery state-of-charge (SOC) and temperature, toggling between bypass mode (SOC >25%) and charging mode (SOC <25%) with Maximum Power Point Tracking (MPPT). Experimental validation on a simulated well platform demonstrated: stable power generation across 25–45 m³/d flow rates, with output voltage and rotation speed exhibiting electromechanical coupling and load-dependent behavior; linear current regulation via PWM control ensuring efficient charging; successful valve-actuated mode switching between generation and bypass states; and continuous charging capability. This system overcomes key limitations of conventional downhole power solutions, providing a reliable, maintenance-free energy source for long-term well monitoring in water injection applications.

Introduction

Driven by the global energy transition and the goals of "carbon peak" and "carbon neutrality," the traditional oil and gas industry is undergoing a green and intelligent transformation. Oilfield intelligence heavily relies on downhole sensor networks to collect key parameters like temperature, pressure, and flow rate in real-time. A new generation of downhole wireless communication and control systems has significantly enhanced data transmission capabilities, achieving millisecond-level response times and centimeter-level positioning accuracy. This system supports real-time monitoring of downhole parameters (e.g., pressure, temperature,

flow rate), remote control of equipment (e.g., intelligent inflow control valves), and data exchange. It is crucial for enhancing recovery rates and reducing the risks associated with manual intervention [1].

However, the widespread adoption of wireless technology faces critical constraints: Devices such as sensors, relay nodes, and actuators require continuous power to maintain functionality, and their energy demands far exceed the capacity of conventional batteries. As well depth increases, wireless signal attenuation intensifies, necessitating the deployment of more relay nodes to boost the signal, further escalating energy consumption [2]. The confined downhole space also limits the installation of high-capacity power supply units [3]. Additionally, the high-temperature, high-pressure (HTHP) environment accelerates battery degradation, leading to increased failure rates [4]. Frequent battery replacements require interrupting oil production operations, causing economic losses.

Downhole power generation technology, overcoming the power supply bottleneck by harvesting energy locally, is therefore key to realizing intelligent oilfields. This technology utilizes energy sources within the downhole environment—such as geothermal gradients, fluid kinetic energy, or chemical energy—converting them directly into electricity underground via energy conversion devices. Its in-situ power supply mode can support high-power-consumption equipment like Measurement While Drilling (MWD) and intelligent completions, representing a vital direction for reducing costs and improving efficiency in oil and gas development. In recent years, research in this field has shown a trend of multidisciplinary integration: breakthroughs in materials science (high-temperature-resistant ceramics and composites enhancing environmental adaptability) and advancements in micro/nano-fabrication technology (MEMS processes enabling device miniaturization) have significantly increased the power density and environmental tolerance of power generation units; intelligent control algorithms (e.g., reinforcement learning-based dynamic energy management) optimize the capture efficiency of intermittent energy sources. Based on energy sources and conversion mechanisms, the main technologies can be divided into three categories:

1. **Fluid Kinetic Energy Harvesting:** Represented by turbine generators, this mature technology offers significant power output. It has achieved commercial application in drilling engineering, with typical examples including Schlumberger's SlimPulse MWD system, PowerDrive rotary steerable tools, and Baker Hughes' LWD wireless logging tools [5][6][7]. Recent research applies turbine technology to wellbore environments; for instance, Arsalan et al.'s turbine system deployed in production wells (250-1500 m³/d) achieved a stable output of 250W [8].
2. **Vibration Energy Harvesting:** Based on piezoelectric or electromagnetic induction principles, this is suitable for dynamic environments like drill string vibration. For example, Trimble's electromagnetic induction device outputs 1.2W under 40Hz vibration [9]; Halliburton's piezoelectric system generates up to 20W using drill pipe rotation [10]; Kim et al.'s miniature piezoelectric MEMS device produces only 100μW, but its compact size offers potential for deployment in narrow wellbores [11].
3. **Thermal and Chemical Energy Conversion:** Includes thermoelectric generators (TEGs) and fuel cells. In oil wells, temperatures are generally constant in horizontal sections, and vertical sections exhibit minimal temperature gradients (approx. 0.025°C/m), both insufficient for power generation via the Seebeck effect, rendering thermal energy harvesting unsuitable downhole [12]. Fuel cells can provide high output power (up to 2kW) through hydrogen-oxygen chemical reactions; however, fuel replenishment downhole is complex and impractical. Furthermore, their output power vastly exceeds the requirements of modern downhole electronics, making this method unsuitable for powering wellbore electronic devices [13][14].

Downhole power generation technology is an essential requirement for oilfield intelligence, but its long-term stable application faces numerous challenges. Firstly, the downhole environment—characterized by

high temperature ($>150^{\circ}\text{C}$), high pressure ($>50\text{ MPa}$), and high corrosivity ($\text{H}_2\text{S}/\text{CO}_2$)—imposes stringent demands on the materials and sealing performance of power generation equipment. This is particularly challenging for traditional turbine technology: turbines are sensitive to fluid cleanliness and flow velocity; fluids containing solids easily cause blade erosion and wear; and the permanent magnets in generators can demagnetize at high temperatures [15][16]. Secondly, the downhole load is complex. The varying operational states of sensors, relay nodes, and actuators at different times cause severe voltage and frequency fluctuations, demanding high stability from the power supply system [17][18]. Thirdly, under normal production conditions, power generation systems are required to operate reliably for years or even decades; failure necessitates a costly workover to pull the tubing for repair, leading to significant economic losses. Finally, the large-scale deployment of downhole power generation systems must also consider economic factors like cost and payback period.

The research and application of downhole power generation technology must first select suitable scenarios. Water injection wells are considered ideal due to the continuous availability of fluid mechanical energy and a relatively simpler downhole environment (fewer impurities). Mature hydraulic turbine technology can be directly applied, but challenges remain: (1) Narrow wellbore space limits turbine size, affecting power output; (2) Power output drops by over 40% under low-flow conditions, potentially preventing the turbine from functioning; (3) Mechanical wear from long-term operation under water flow impact can lead to equipment failure [19][20]. To address these issues, this study proposes a downhole compact magnetically levitated turbine generator technology suitable for water injection wells. Based on turbine principles, this technology integrates the generator with a secondary battery, optimizing flow path design and power management according to the hydrodynamics and spatial constraints of injection wells. It aims to effectively reduce mechanical losses, enhance power generation efficiency, and achieve "on-demand" and "intermittent" power generation.

Structure and Principle

The power generation system described herein employs a turbine-type generator structure, as shown in Fig. 1. It consists of a drive end, a power generation end, and a magnetic coupling shaft connecting them. Downhole fluid enters the device through tangential inlet ports and impacts the impeller, causing it to rotate. The turbine drives the rotation of the outer magnetic rotor. Simultaneously, under the action of magnetic drive, the inner magnetic rotor synchronously follows the rotation of the outer magnetic rotor. Consequently, the generator rotor, connected to the inner magnetic rotor, also rotates. This rotation interacts with the generator stator to produce electrical energy. The drive end incorporates a magnetic bearing structure. This enables the impeller and its connected dual half-shafts (collectively referred to as the rotor) to achieve magnetic levitation within the generator cavity. Axial positioning is maintained solely via a single contact point at the end. This design significantly reduces frictional resistance caused by rotor rotation. Consequently, the generator's minimum startup flow rate requirement is substantially lowered.

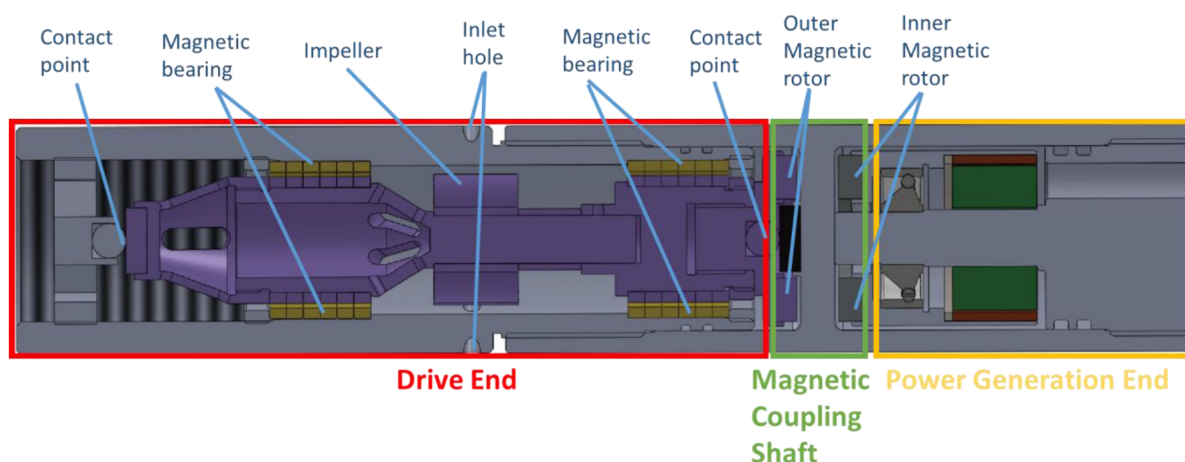


Figure 1—Structure of a magnetic levitation turbine power generation system

This generator is positioned eccentrically within the wellbore flow path and does not occupy the central channel. Electricity is generated only when water flows through the eccentric channel; the generator remains inactive during flow through the central channel. A key system component is the valve, which directs water flow and thereby determines generator operation. The flow channel structure of the intermittent power generation system is shown in Fig. 2.

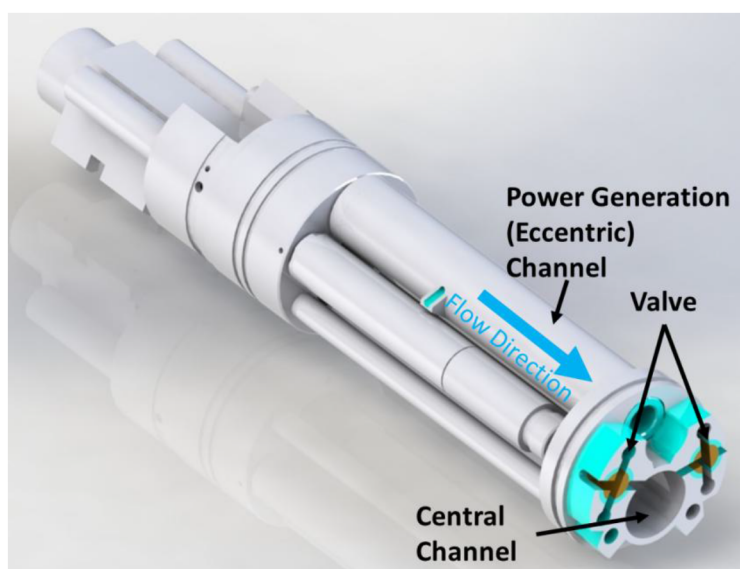


Figure 2—Flow channel structure of the intermittent power generation system

The power generation system controls the valve's rotation angle to select the flow path based on power generation requirements. A schematic diagram illustrating the flow paths connected by the valve is shown in Fig. 3. The initial state, where the valve fully connects the power generation channel, is defined as the 0° angular position. When the valve rotates counterclockwise, the resulting flow path configuration and the consequent operational status of the system are summarized in Table 1.

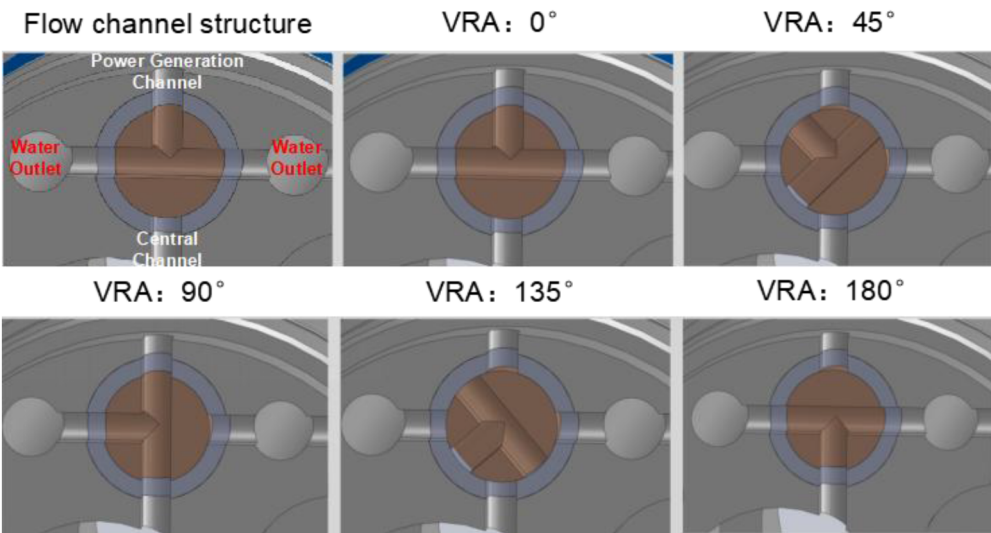


Figure 3—Flow Path Connection Status

Table 1—Valve Rotation Angle vs. Flow Path Connection Status

Valve Rotation Angle (CCW)	Flow Path Connection Status	Power Generation System Status
0°	Power channel independently connected	Power generation mode
45°	Both flow paths blocked	Zero flow
90°	Power channel parallel-connected with central channel	Bypass mode
135°	Both flow paths blocked	Zero flow
180°	Central channel independently connected	Bypass mode

To provide stable power supply to downhole electronics, this power generation system is equipped with a rechargeable battery. The generator output first charges the rechargeable battery, and then the battery supplies power to the electronics. This design suppresses the impact of voltage fluctuations induced by flow variations on the electronics and reduces the demands on the rectification circuit. The main control chip, serving as the system core, monitors the battery status and governs the flow path selection valve. Its control logic is illustrated in Fig. 4.

During normal operation, the main control chip first initializes the system and monitors the battery state of charge (SOC). When the SOC exceeds a preset threshold (e.g., 25%), the valve maintains connection to the central channel, placing the system in bypass mode where the generator remains inactive. If the SOC falls below the threshold, the main control chip actuates the valve to connect the power generation channel, switching the system to charging mode. In this mode, Maximum Power Point Tracking (MPPT) control is implemented, and charging continues until the SOC surpasses 95%. At this point, the main control chip readjusts the system to bypass mode. Throughout this process, the main control chip continuously monitors the rechargeable battery temperature. Charging proceeds normally while the temperature remains below 150°C. Should the temperature exceed 150°C, charging is terminated immediately and the system reverts to bypass mode. In bypass mode, the main control chip persistently monitors the battery SOC to facilitate the charging cycle described above. Simultaneously, the main control chip monitors flow conditions. If no flow is detected for 30 consecutive minutes, the system enters sleep mode. During sleep mode, the main control chip periodically checks the flow rate at predetermined intervals. Once the flow reaches the threshold required for power generation, the system resumes monitoring the SOC and reactivates power generation.

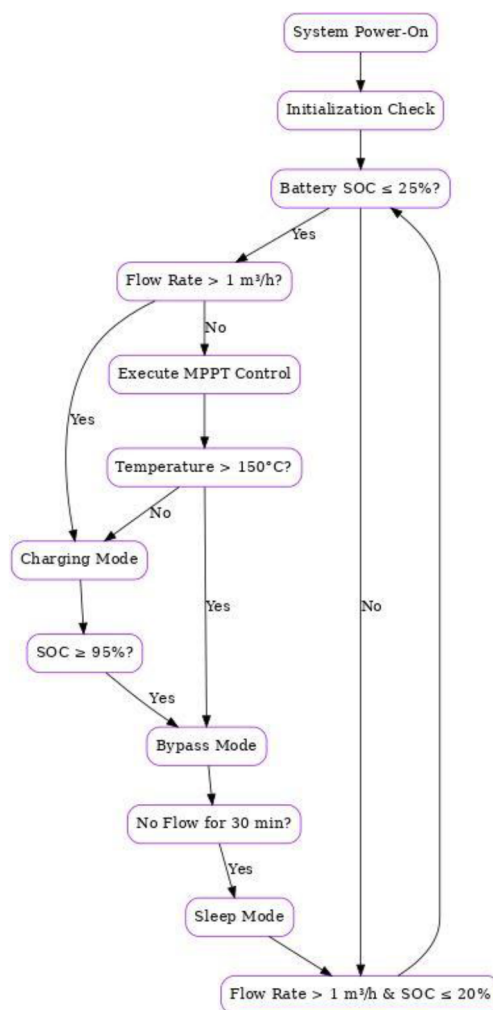


Figure 4—Working principle of the intermittent controllable power generation system

Experiment

The assembled prototype is shown in Fig. 5. The prototype is connected to a simulated water injection well platform, which comprises three main components: a power system, a pipeline system, and a control system. This platform enables precise simulation of downhole water injection conditions. The power system features a plunger pump rated at 150 m³/d and 4 MPa. This pump, coupled with a 0-50 Hz variable frequency drive (VFD) and a 7.5 kW VFD control cabinet (AC 380 V power supply), enables precise flow rate regulation from 0 to 150 m³/d. A 2 m³ stainless-steel water tank (316L material) ensures circulation stability. The pipeline system incorporates an electromagnetic flowmeter ($\pm 0.5\%$ accuracy, 3-280 m³/d range) and an electric control valve (0.1% resolution). An electric on/off valve with a 15-second response time simulates the wellbore scenario. The control system integrates an industrial computer (AC 220 V power supply) and a data acquisition module. Integrated software facilitates remote pump start/stop, flow rate adjustment, and real-time data monitoring.

The start-up flow rate of the generator was tested first. Under no-load conditions, the prototype generator effectively started when the water injection flow rate reached the critical value of 24.5 m³/d. Given that the conventional water injection flow rates in oilfield wells typically exceed 24.5 m³/d, this threshold design satisfies actual production requirements. Using 24.5 m³/d as the baseline, experiments tested the power generation and charging performance at flow rates of 25, 30, 35, 40, and 45 m³/d. The performance of the flow path selector valve was also tested. The results and analysis are as follows:



Figure 5—Prototype of the intermittent controllable power generation system

Characteristics of Generator Speed and Output Voltage versus Charging PWM Duty Cycle

Figs 6 and 7 illustrate the variation of rotational speed of the generator and output voltage with the charging PWM duty cycle under multiple flow rates, revealing the following:

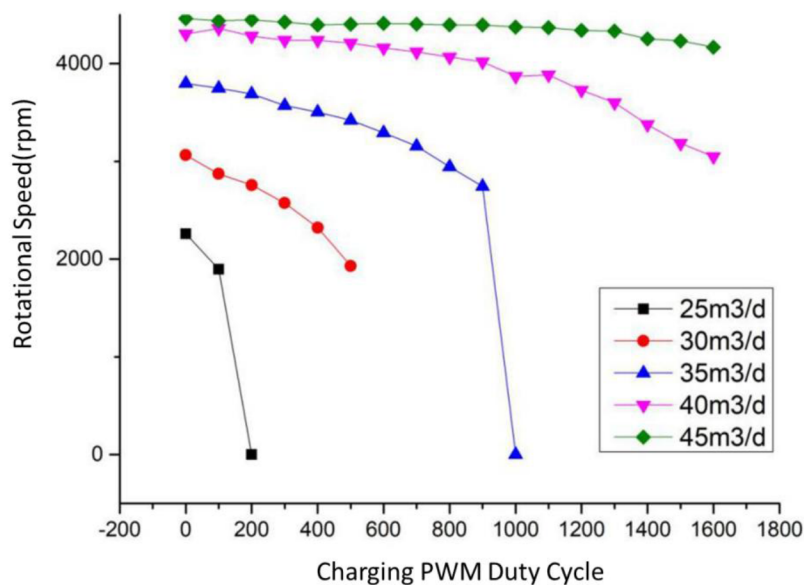


Figure 6—Generator speed versus charging PWM duty cycle

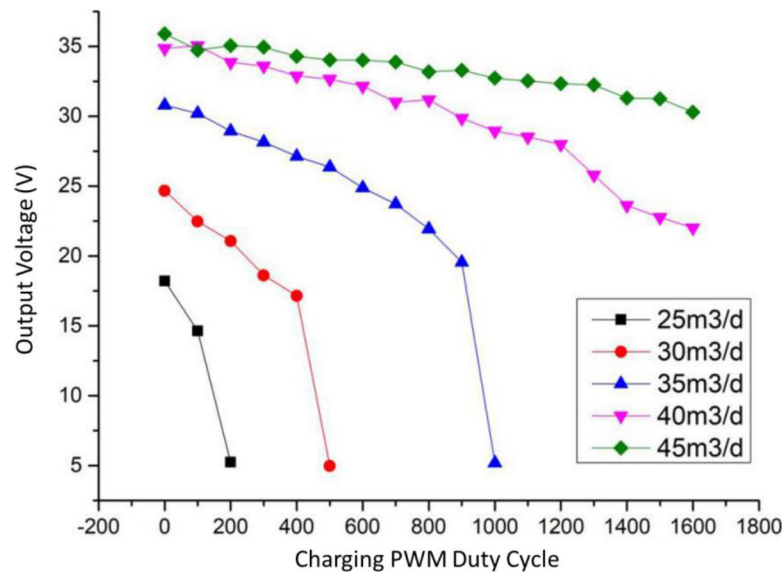


Figure 7—Generator output voltage versus charging PWM duty cycle

- Electromechanical Parameter Coupling:** Generator speed and output voltage exhibit significant synchronous response to changes in the PWM duty cycle, verifying the electromechanical coupling relationship between speed and voltage. This mechanism originates from the proportional relationship between electromotive force and angular velocity in the law of electromagnetic induction.
- Load Effect:** Under constant flow rate conditions, generator speed and voltage decrease as the PWM duty cycle increases. This phenomenon is attributed to increased energy dissipation caused by rising load impedance, resulting in reduced efficiency in converting rotor kinetic energy into electrical energy.
- Critical Load Threshold:** A critical load threshold (PWM_max) exists across all flow rate conditions; exceeding this value causes system instability and shutdown. Experiments indicate that PWM_max is positively correlated with flow rate, its dynamic essence being the process of balancing fluid kinetic energy and electromagnetic torque.
- Flow Rate-Load Sensitivity Variation:** The impact of PWM adjustment on speed/voltage is lower at high flow rates ($>40 \text{ m}^3/\text{d}$) compared to low flow rates, indicating that enhanced system inertia significantly suppresses load disturbance sensitivity. This characteristic has important engineering implications for robust system design: multi-parameter coordinated control strategies are required in high-flow wellbore regions to achieve stable output.

Characteristics of Charging Current and Power versus Charging PWM Duty Cycle

Figures 8 and 9 reveal the variation of charging current and power with the charging PWM duty cycle under multiple flow rates, showing:

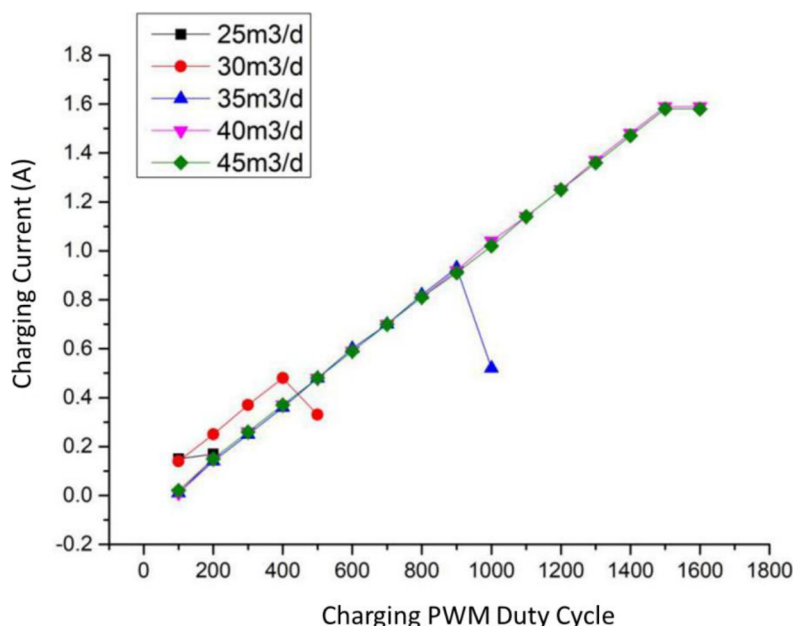


Figure 8—Charging current versus PWM duty cycle

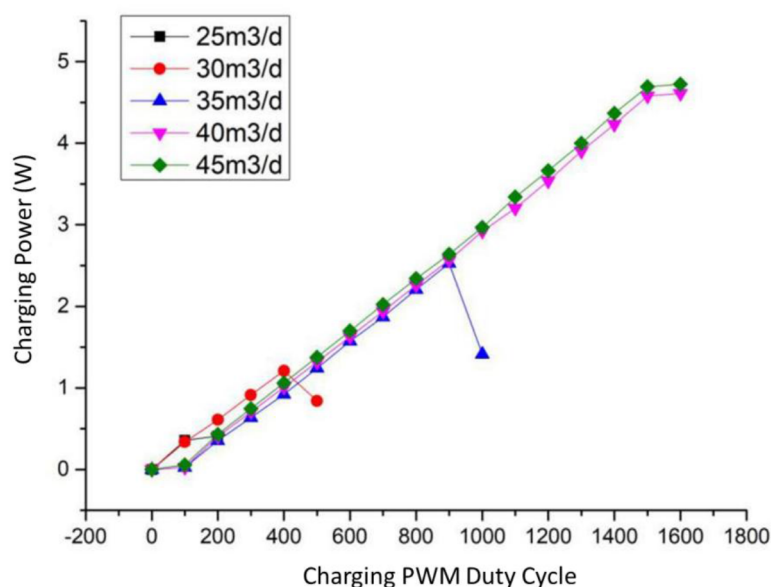


Figure 9—Charging power versus PWM duty cycle

- Critical Load Threshold Consistency:** A critical load threshold (PWM_max) was observed under all flow rate conditions. Its value matches the speed/voltage instability threshold shown in Figs. 6/7, verifying the unified instability boundary of the electromechanical system.
- Charging Parameter Regulation:** At a constant PWM duty cycle, charging current and power exhibit flow rate independence. This phenomenon stems from the active regulation of input power by the closed-loop feedback mechanism of the rectifier circuit. Experimental data indicates a strictly linear relationship between charging current and PWM (e.g., at 45 m³/d, $R^2=0.9999$ for $200 \leq \text{PWM} \leq 1500$, see Fig. 10). This characteristic confirms the excellent output stability and voltage fluctuation suppression capability of the power management module, meeting the requirements for efficient energy storage unit charging.

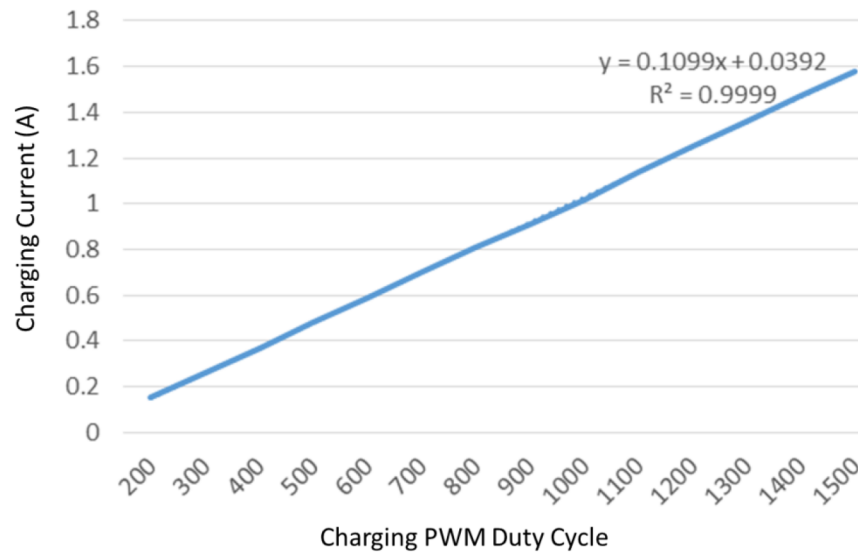


Figure 10—Charging current trend versus PWM duty cycle at 45 m³/d

Continuous Charging Test

A battery charging test was conducted at 32 m³/d for 6 hours, during which the voltage increased from 2.6 V to 3.2 V. Approximately 7.4 Wh of energy was charged, as shown in Fig. 11. The test reveals:

- The battery voltage exhibits a typical non-linear growth trend, with a significantly higher initial rise rate compared to the later saturation phase.
- The charging current fluctuation amplitude throughout the cycle is $\leq \pm 1.5\%$, confirming the excellent constant current control capability of the rectifier circuit.
- Generator speed shows a weak negative correlation with increasing battery voltage, likely caused by electromechanical coupling effects resulting from changes in load impedance characteristics.

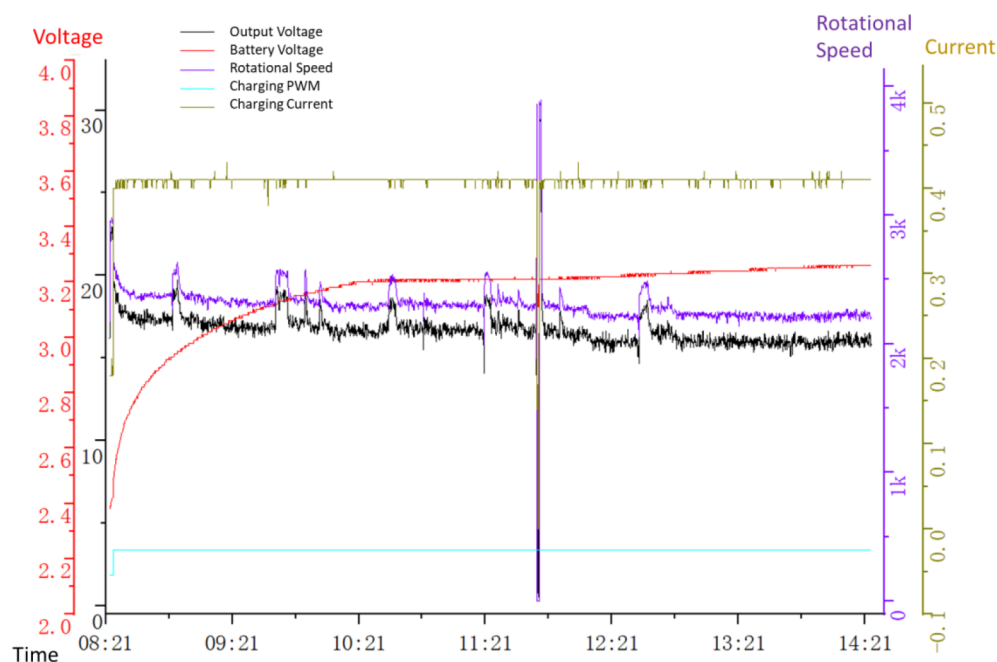


Figure 11—Long-term generator charging test (Flow rate: 32 m³/d, Charging PWM: 400)

Based on measured charging power data, Table 2 presents the minimum theoretical charging duration for a fully discharged (to 3.6 V cutoff) 2Ah 18650 LiFePO4 battery at room temperature (assuming 90% charging efficiency). In practical applications, high-temperature conditions (70°C) cause approximately 30% degradation in effective battery capacity. Furthermore, system energy conversion efficiency exhibits a negative correlation with increasing output power. Consequently, the actual operating duration is reduced by approximately 40% compared to the theoretical value, necessitating correction using temperature compensation coefficients.

Table 2—Battery charging duration under different flow rates

Flow Rate (m³/d)	Maximum Charging Current (A)	Charging Duration
25	0.15	14h49min
30	0.48	4h38min
35	0.93	2h23min
40	1.59	1h24min
45	1.59	1h24min

Flow Path Switching Test

Systematic testing of the flow path selection valve's operational characteristics was conducted under a constant flow rate of 30 m³/d. Prior to the test commencement, the valve was in the closed position. Following test initiation, the valve was rotated incrementally by 45° every 2 minutes, starting from the 0° position (the definition of valve rotation angle is shown in Fig. 3. The dynamic response characteristics of the generator's rotational speed and output voltage over time were recorded by the monitoring system (Fig. 12).

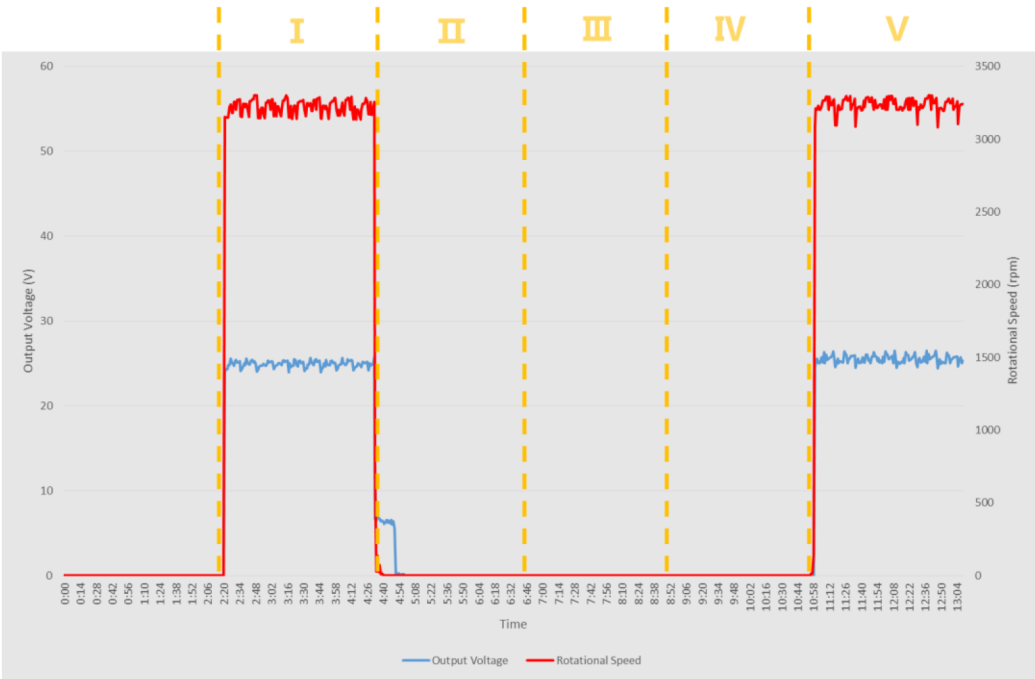


Figure 12—Valve actuation response curves

The operational characteristics corresponding to the valve actuation stages depicted in Fig. 12 are as follows:

Stage I: Valve angle 0° , power generation channel open. Generator rotational speed stabilized near 3000 r/min, with the output voltage stable at 25 V.

Stage II: Valve angle 45° , flow path fully closed. Generator rotational speed and output voltage plummeted to zero.

Stage III: Valve angle 90° , both the power generation channel and the central channel connected to the outlet, resulting in the power generation channel being hydraulically short-circuited. Generator rotational speed and output voltage remained at zero.

Stage IV: Valve angle 135° , flow path fully closed. Generator rotational speed and output voltage remained at zero.

Stage V: Power generation channel reopened. Rotational speed and voltage rapidly recovered to the levels observed in Stage I.

This experiment validates the effective control of generator operation through flow path selection. The rapid valve response demonstrates the feasibility of an intermittent power generation mode.

Conclusion

This paper demonstrates a breakthrough intermittently controllable magnetically levitated turbine generator system that effectively addresses the sustainable power challenge for downhole electronic devices in water injection wells. By integrating magnetic levitation to eliminate mechanical wear and enable ultra-low startup flow rates ($24.5 \text{ m}^3/\text{d}$), an eccentric flow path selector valve for on-demand power generation/bypass switching, and intelligent energy management with SOC-based mode control and MPPT charging, the system overcomes critical limitations of conventional downhole turbines. Experimental validation confirmed stable operation across $25\text{--}45 \text{ m}^3/\text{d}$ flow rates, precise PWM current regulation, reliable valve-actuated mode transitions, and continuous charging capability (7.4 Wh in 6 hours). This maintenance-free solution ensures long-term power autonomy for downhole monitoring, eliminating battery replacement needs and supporting intelligent oilfield development.

Further research should prioritize enhancing high-temperature resilience ($>150^\circ\text{C}$) for harsh well environments through advanced materials, exploring hybrid energy harvesting (e.g., combining fluid kinetic and vibration energy) to ensure reliability during low-flow periods, and optimizing system miniaturization via MEMS technology for narrower wellbores. Long-term field validation under corrosive conditions and dynamic flow regimes is essential to assess durability. Additionally, integrating AI-driven predictive control could dynamically optimize charging thresholds based on historical flow data and real-time energy demand, while digital twin integration would enable proactive maintenance. These advancements will solidify the system's role in enabling autonomous, low-carbon oilfield operations and scalable real-time reservoir management.

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